



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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Agenda

- Chapter 7 Operations Planning and Control

Demand Forecasting

- Forecasting assumes a causal system in that future is somehow related to the past in someway
- Forecasts can provide estimated values because of randomness (uncertainty, risk or simply unknown)
- Forecasts may be more accurate for groups vs. individuals
- Forecast accuracy decreases as time horizon increases

Demand Forecasting

- Types of forecast:
 - Judgmental - uses subjective inputs
 - Time series - uses historical data assuming the future will be like the past

Judgmental Forecasts

- Executive opinions
- Customer contact
- Consumer surveys
- Outside opinion
- Opinions of managers and staff



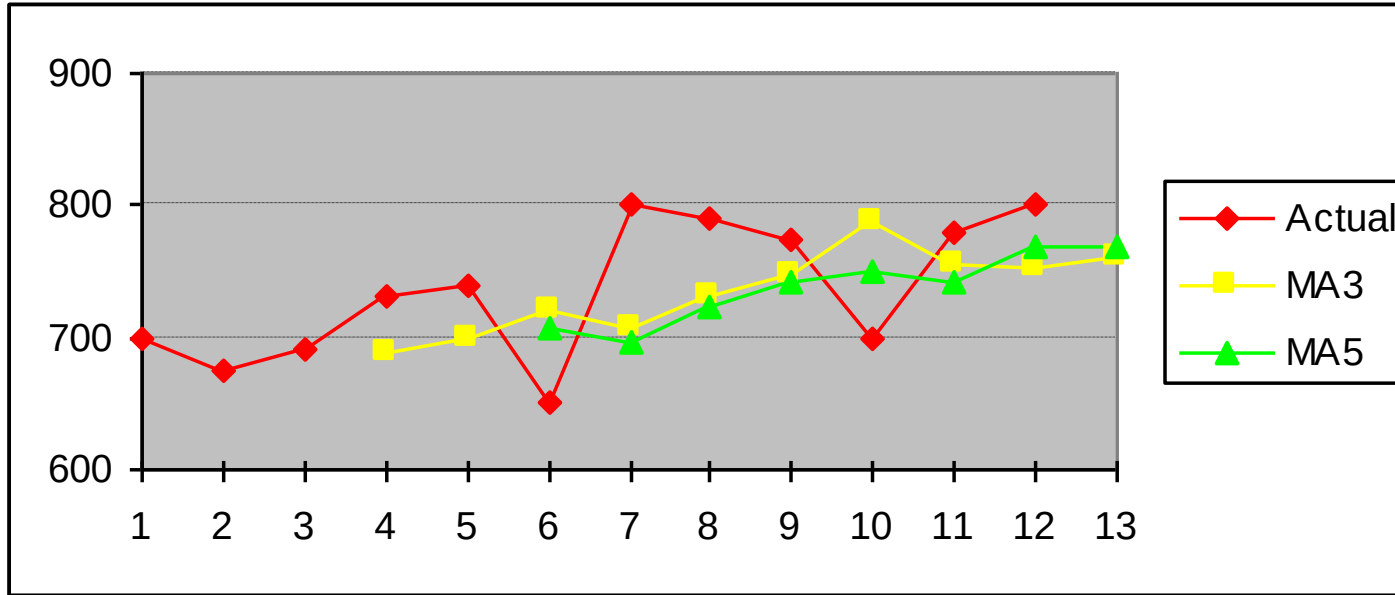
Time Series Forecasts

- Trend - long-term movement in data
- Seasonality - short-term regular variations in data
- Irregular variations - caused by unusual circumstances
- Random variations - caused by chance

Elementary techniques of demand forecasting

- Moving Averages
- Exponentially weighted moving average
- Linear regression

Simple Moving Average



The next period's demand: the average of the actual demand values for the last n time periods

$$\hat{x}_t = \frac{\sum_{i=1}^n x_{t-i}}{n}$$

Number of time periods to include

Example: Compute a 3-period moving average for product A

Period	Age	Demand
1	5	42
2	4	40
3	3	43
4	2	40
5	1	41

$$\hat{x}_4 = \frac{43 + 40 + 42}{3} = 41.6$$

$$\hat{x}_5 = \frac{40 + 43 + 40}{3} = 41$$

$$\hat{x}_6 = \frac{41 + 40 + 43}{3} = 41.3$$

Weighted moving average

$$\hat{x}_t = \frac{\sum_{i=1}^n c_i x_{t-i}}{n}$$

$$n = \sum_{i=1}^n c_i$$

Exponentially weighted moving average (EWMA)

- Premise-The most recent observations might have the highest predictive value
 - We should give more weight to the more recent time periods when forecasting
 - Assigning weights to the demand value of previous periods in inverse proportion to their age
 - Last period's forecast, last period's actual demand, smoothing constant

Next forecast = Previous forecast + α (Actual – Previous Forecast)

Example: $\alpha=20\%$

The most recent observation (short-term trend) contributes 20% weight, and the remaining 80% weight is distributed (exponentially decaying) across all earlier observations (long-term trend)

$$\hat{x}_t = \hat{x}_{t-1} + \alpha(x_{t-1} - \hat{x}_{t-1})$$

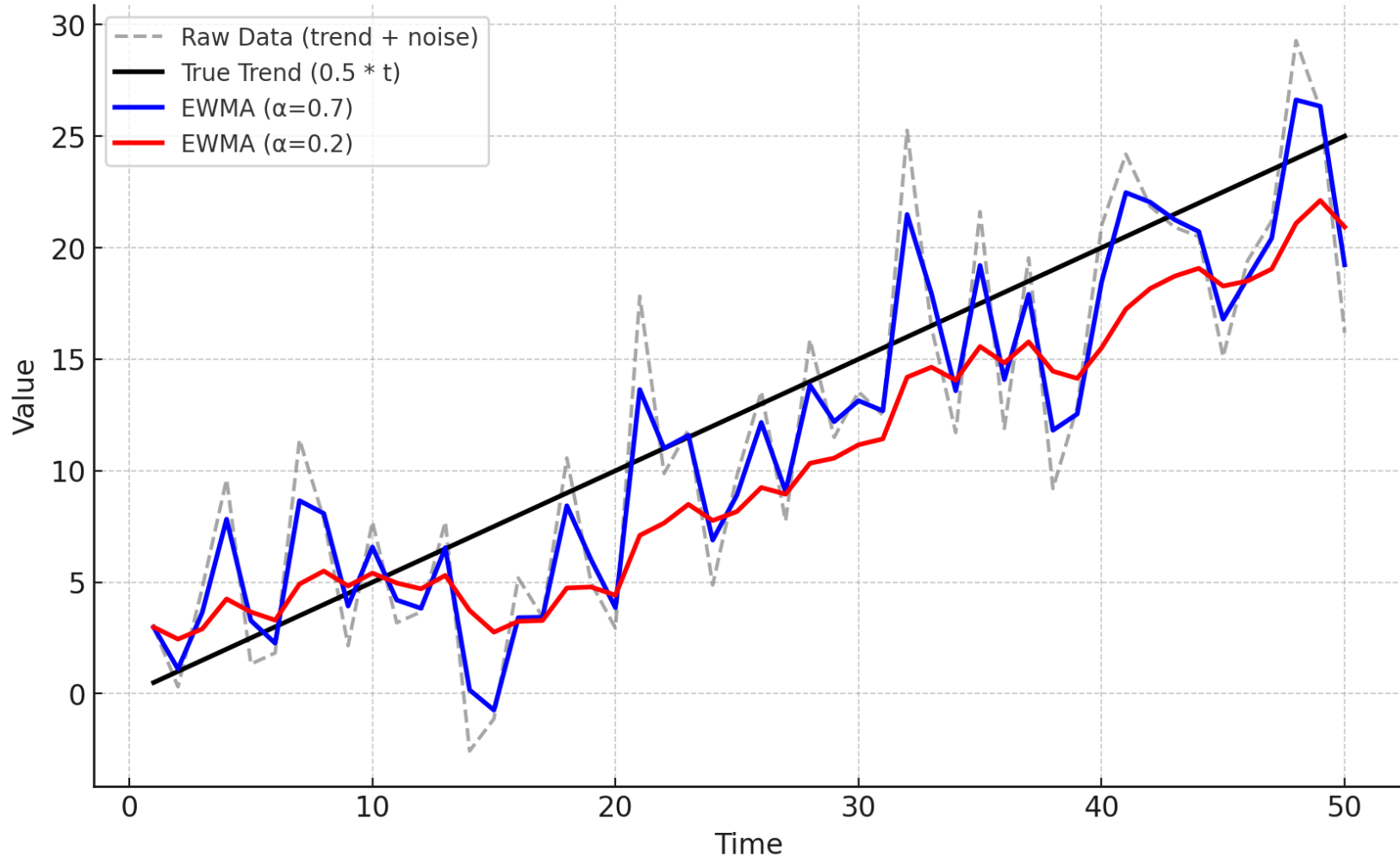
$$\hat{x}_t = \alpha x_{t-1} + (1 - \alpha)\hat{x}_{t-1}$$

Example: Prepare a forecast using a smoothing constant of 0.40

Period	Number of complaints	Forecast	Calculations
1	60		
2	65	60	Use previous value of series
3	55	62	$60 + 0.4(65-60)=62$
4	58	59.2	$62 + 0.4(55-62)=59.2$
5	64	58.72	$59.2 + 0.4(58-59.2)=58.72$
		60.83	$58.72 + 0.4(64-58.72)=60.83$

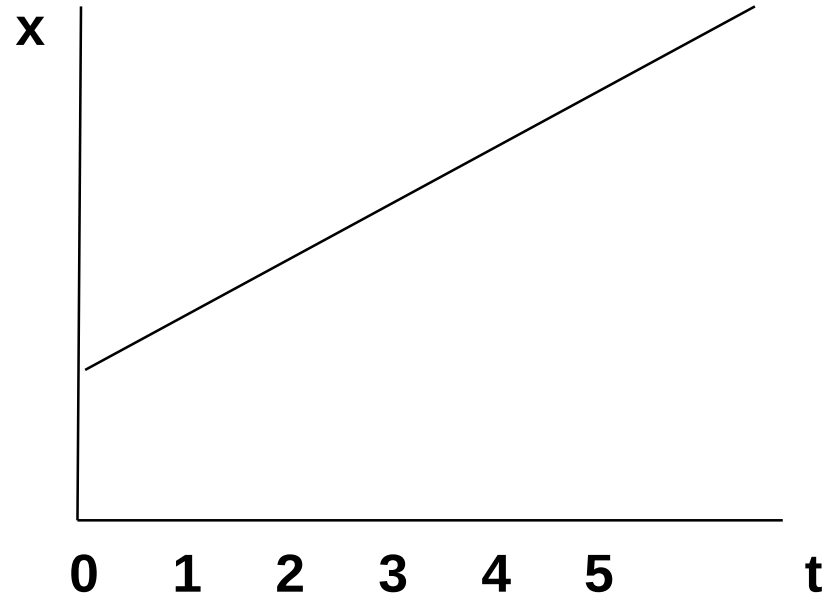
Impact of the smoothing factor on the forecast

EWMA Smoothing with Different α Values



Linear regression equation

$$\hat{x}_t = a + bt$$

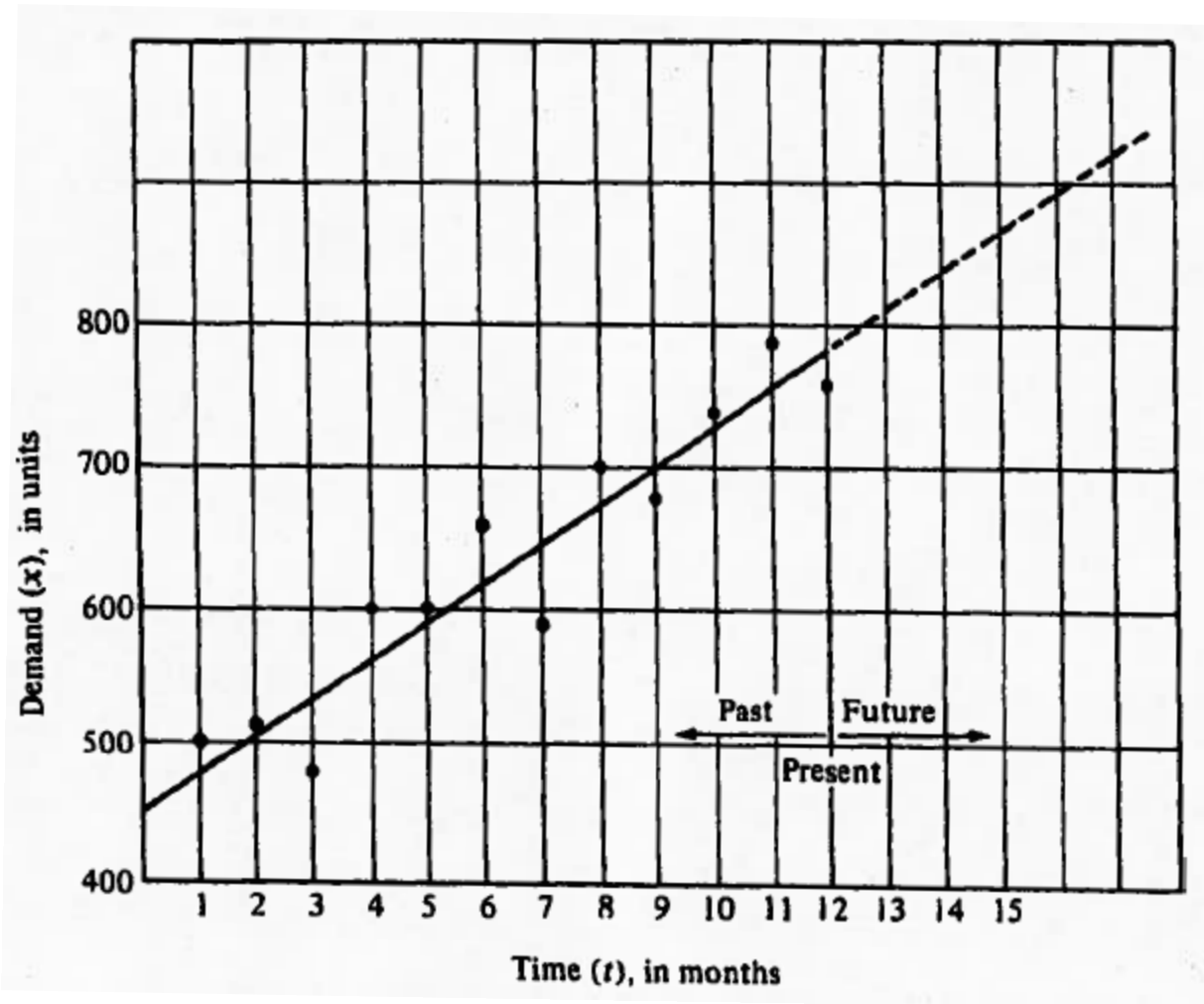


- a is the value of X at t=0
- b is similar to the slope. However, since it is calculated with the variability of the data in mind, its formulation is not as straight-forward as our usual notion of slope

Linear regression equation

$$x_t = a + bt$$

Month	Demand
Jan,	500
Feb.	510
Mar.	480
Apr.	600
May	600
June	660
July	590
Aug.	700
Sept.	680
Oct.	740
Nov.	790
Dec.	760



Calculating a and b

$$b = \frac{n \sum_{i=1}^n t_i x_i - \sum_{i=1}^n t_i \sum_{i=1}^n x_i}{n \sum_{i=1}^n t_i^2 - (\sum_{i=1}^n t_i)^2}$$

$$a = \frac{\sum_{i=1}^n x_i - b \sum_{i=1}^n t_i}{n}$$

Linear regression example

t Week	t^2	x Sales	tx
1	1	150	150
2	4	157	314
3	9	162	486
4	16	166	664
5	25	177	885
$\Sigma t = 15$ $(\Sigma t)^2 = 225$	$\Sigma t^2 = 55$	$\Sigma x = 812$	$\Sigma tx = 2499$

Linear regression calculation

$$b = \frac{5(2499) - 15(812)}{5(55) - 225} = \frac{12495 - 12180}{275 - 225} = 6.3$$

$$a = \frac{812 - 6.3(15)}{5} = 143.5$$

$$x = 143.5 + 6.3t$$